

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2019

Fraser 138 kV
December 20, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2019. The Interconnection Customer proposes a 170.9 MW solar photovoltaic facility interconnecting at the Fraser 138 kV substation in the West Virginia area. The request was received on February 26, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 516 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$762,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Lochmere Hydroelectric Plant – Hinton Ap Substation 138kV
Contingency: Manatee – Hinton Ap Substation 345kV L
Loading: 109.2% (Rating: 265 MVA)
DFAX: 35.5%
Pre-project loading: 86.3%
2. Facility: Lochmere Hydroelectric Plant – Pitchgen 138kV Line
Contingency: Hinton Ap – Manatee 345kV line
Loading: 108.6% (Rating: 272 MVA)
DFAX: 35.5%
Pre-project loading: 86.3%
3. Facility: Briar – Pitchgen 138kV L
Contingency: Manatee – Hinton Ap 345kV
Loading: 107.9% (Rating: 419 MVA)
DFAX: 52.9%
Pre-project loading: 86.3%
4. Facility: Briar – Fraser 138kV line
Contingency: Manatee – Hinton Ap 345kV Line
Loading: 107.7% (Rating: 422 MVA)
DFAX: 52.9%
Pre-project loading: 86.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lochmere Hydroelectric Plant – Hinton Ap Substation 138kV	Reconductor line	\$1,388,000	18% (\$246,000)
Pitchgen – Lochmere Hydroelectric Plant 138kV	Reconductor line	\$742,000	18% (\$132,000)
Briar – Pitchgen 138kV L	Reconductor line	\$1,278,000	17% (\$221,000)
Briar – Fraser 138kV line	Reconductor line	\$947,000	17% (\$163,000)
	Total allocated to project		\$762,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.